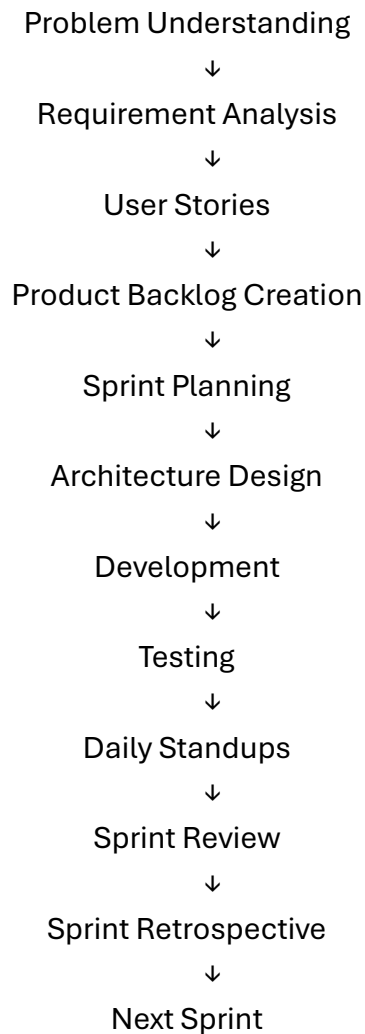


## Complete Agile Workflow for Your School Management System

The project lifecycle should look like this:



This cycle repeats until the product is complete.

---

### 1. Requirement Analysis

First step is **understanding the system properly**.

You already have the **problem statement**.

Now extract **functional requirements**.

Example:

| Requirement ID | Requirement                            |
|----------------|----------------------------------------|
| R1             | Users should be able to register       |
| R2             | Admin should manage students           |
| R3             | Teacher should mark attendance         |
| R4             | Students should access study materials |
| R5             | Parents should view grades             |
| R6             | Admin should generate reports          |

Also define **Non Functional Requirements**

Examples:

- Security
- Performance
- Scalability
- Data privacy

Example:

System should support 1000 users

Login should be secure

Data should be encrypted

---

## 2. Identify Epics

An **Epic = Large Feature**

For your system, epics could be:

| Epic                  |
|-----------------------|
| User Management       |
| Class Management      |
| Attendance Management |

|                       |
|-----------------------|
| <b>Epic</b>           |
| Exam & Grades         |
| Fee Management        |
| Communication         |
| Educational Resources |
| Reports & Analytics   |

These epics will later be split into **user stories**.

---

### 3. Writing User Stories

User stories follow this format:

As a <user>  
I want to <action>  
So that <benefit>

Examples:

#### User Story 1

As an administrator  
I want to add new students  
So that student records can be maintained

#### User Story 2

As a teacher  
I want to mark attendance  
So that student attendance can be tracked

#### User Story 3

As a parent  
I want to view my child's grades  
So that I can monitor academic performance

Each story should include **acceptance criteria**.

Example:

Teacher can mark attendance  
Attendance saved in database  
Attendance report generated

---

#### 4. Product Backlog Creation (Jira)

All user stories go into **Product Backlog**.

Example backlog:

| Story               | Priority |
|---------------------|----------|
| User Registration   | High     |
| Login System        | High     |
| Student Management  | High     |
| Attendance Tracking | Medium   |
| Exam Management     | Medium   |
| Event Management    | Low      |

The **Product Owner** **prioritizes** the backlog.

---

#### 5. Estimation Using Planning Poker

Each story is estimated using **Story Points**.

Typical scale:

- 1 → Very easy
- 2 → Easy
- 3 → Medium
- 5 → Complex
- 8 → Very complex
- 13 → Extremely complex

Example:

| Story | Story Points |
|-------|--------------|
| Login | 3            |

| Story                | Story Points |
|----------------------|--------------|
| Student Registration | 5            |
| Attendance Module    | 8            |
| Exam System          | 13           |

Planning poker ensures **team agreement**.

---

## 6. Sprint Planning

A **Sprint** = fixed development cycle.

Usually:

Sprint Duration = 2 weeks

Example Sprint 1:

### Task

User registration

Login system

Basic dashboard

Example Sprint 2:

### Task

Student management

Teacher assignment

Class schedule

Example Sprint 3:

### Task

Attendance module

Reports

Sprint planning decides:

What will be done in this sprint  
Who will do what

---

## 7. Jira Setup (Important)

Create a **Scrum project in Jira**.

Then create:

Epic

User Stories

Tasks

Bugs

Typical Jira hierarchy:

Epic

└─ User Story

    └─ Task

Example:

Epic: Attendance Management

User Story:

Teacher marks attendance

Tasks:

- Create attendance UI
  - Create backend API
  - Store attendance in database
  - Generate report
- 

## 8. Architecture Design

Before coding, define system architecture.

Example for your project:

### Frontend

React

HTML

CSS

JavaScript

## Backend

Node.js / Spring Boot / Django

REST APIs

## Database

MySQL

MongoDB

Example architecture:

Frontend (React)

↓

REST API

↓

Backend (Node/Spring)

↓

Database

---

## 9. Development

Develop features according to sprint tasks.

Workflow:

Create branch

Develop feature

Commit code

Push to Git

Create Pull Request

Code review

Merge

Example Git workflow:

main

├─ feature-login

├─ feature-attendance

└─ feature-student-management

---

## 10. Testing

Testing should happen continuously.

Types of testing:

### **Unit Testing**

Test individual components.

Example:

Login API test

Attendance function test

### **Integration Testing**

Test module interactions.

Example:

Attendance + Student database

### **System Testing**

Test entire application.

Example:

Full school workflow

---

## **11. Daily Standup Meeting**

Every day (15 minutes).

Each member answers **3 questions**:

What did I do yesterday?

What will I do today?

Any blockers?

Example:

Yesterday:

Completed login API

Today:

Working on registration page

Blocker:

Database connection issue

---



## 12. Sprint Review

At the end of the sprint.

Team **demonstrates the product**.

Example:

Sprint 1 Demo

✓ User registration

✓ Login

✓ Dashboard

Stakeholders give feedback.

---

## 13. Sprint Retrospective

This is where teams **improve their process**.

Questions:

What went well?

What went wrong?

What should we improve?

Example:

Good:

Team collaboration

Problem:

Late code merges

Improvement:

Use smaller commits

---

## 14. Repeat the Cycle

Agile is **iterative**.

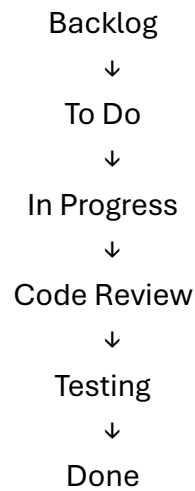
Sprint 1 → Sprint 2 → Sprint 3 → Sprint 4

Each sprint adds new features.

---

## How Jira Board Should Look

Typical Scrum board:



Example task movement:

Create Login Page

Backlog → To Do → In Progress → Done

---

## Roles in Your Team

You should assign clear roles.

| Role          | Responsibility            |
|---------------|---------------------------|
| Product Owner | Manages backlog           |
| Scrum Master  | Facilitates agile process |
| Developers    | Build features            |
| Testers       | Verify system             |

In small college teams, people often **play multiple roles**.

---

## Realistic Sprint Plan for Your Project

Example **4 sprint structure**.

### Sprint 1

User registration  
Login system  
Role management

### **Sprint 2**

Student management  
Class management  
Teacher assignment

### **Sprint 3**

Attendance system  
Educational resources

### **Sprint 4**

Exam system  
Reports  
Notifications